

Static Timing Analysis

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Course Overview

1. STA Concepts
 - 1 lecture
2. Delay Modeling
 - 1 lecture
3. Interconnect Parasitics
 - 1 lecture
4. Delay Calculation
 - 1 lecture
5. **Configuring the STA Environment**
 - 1 lecture
6. Timing Checks
 - 1 lecture
7. Crosstalk and Noise
 - 1 lecture

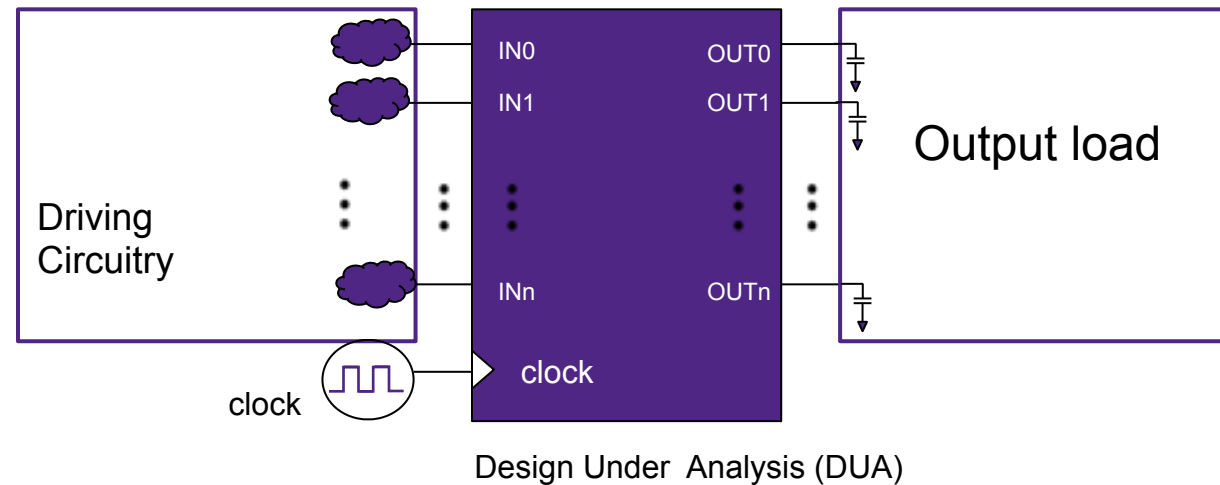
5. Configuring the STA Environment

Getting Ready for STA Tool Usage

- To identify all the timing issues in the gate level design, the following information is required:
 - Clocks
 - I/O timing characteristics
 - False paths
 - Multicycle paths
 - Environment

Setting Design Environment

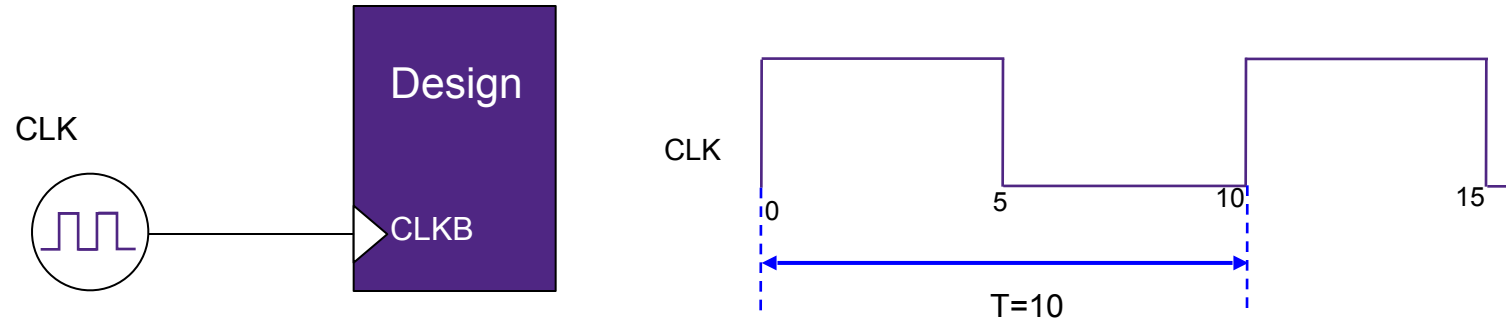
- To perform STA on this design, it is required to:
 - Specify the clocks
 - Set timing constraints for input and output paths
 - Define driving circuitry
 - Set output loads



Basic Clock Details

- Clock Definition Components:
 - Source
 - ♦ Port of a design
 - Period
 - ♦ Time period of the clock (frequency)
 - Duty cycle
 - ♦ Positive and negative phase
 - Edge times
 - ♦ Rising edge and falling edge
 - Variability
 - ♦ Changes in frequency

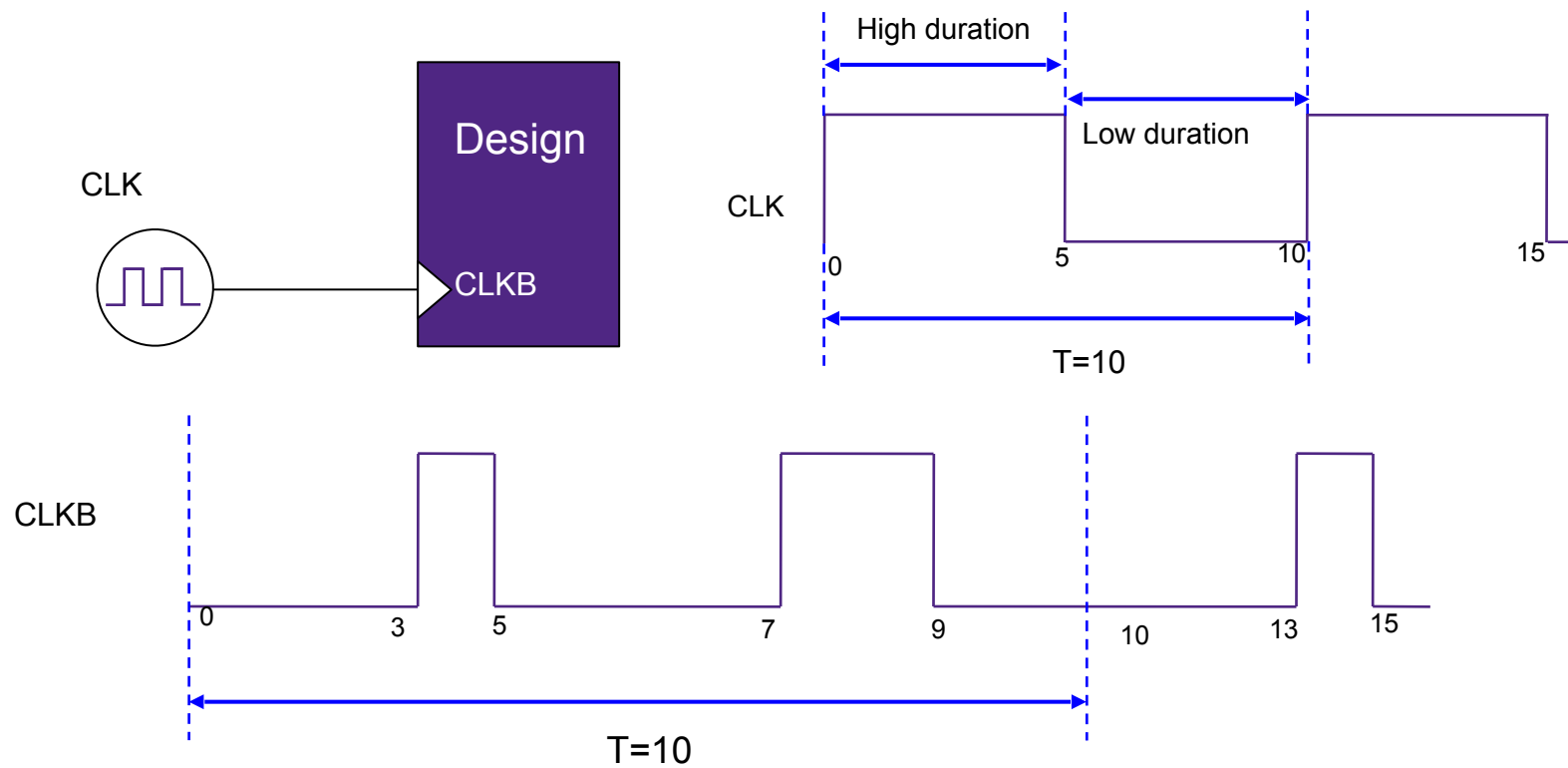
Clock Definition



```
create_clock -name CLK -period 10 [get_port CLKB]
```

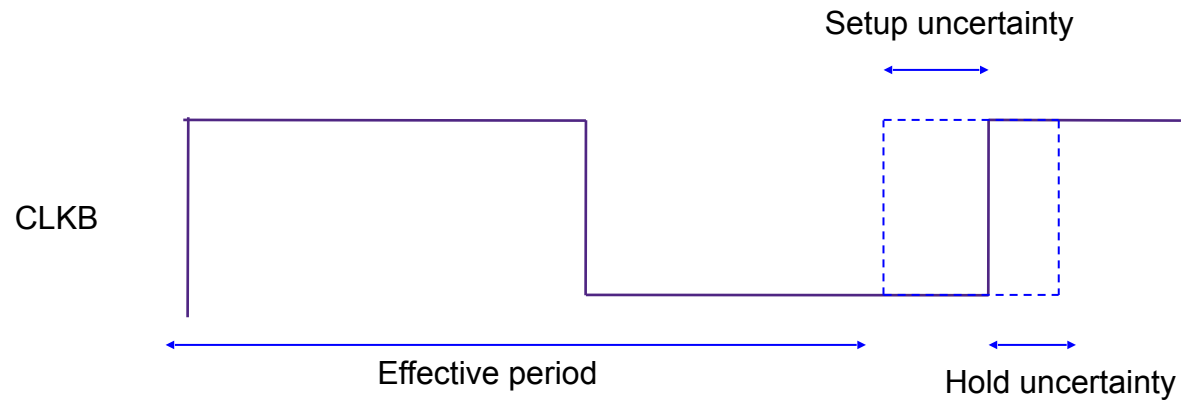

Clock Waveform

```
create_clock -period 10 \  
             -waveform {3 5 7 9} [get_port CLKB]
```



Clock Uncertainty

```
set_clock_uncertainty -setup 0.4 [get_clocks CLKB]  
set_clock_uncertainty -hold 0.04 [get_clocks CLKB]
```



Uncertainty= clock skew + clock jitter

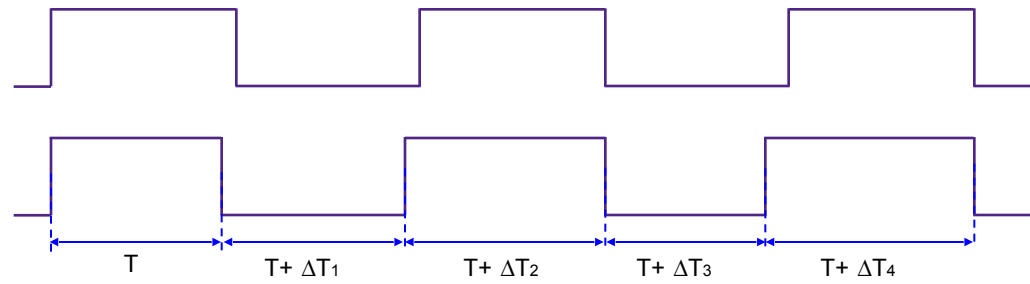
Clock Skew Types

- Global
 - Clock difference between any pair of sequential elements of the whole design
- Local
 - Clock difference between any pair of sequential elements that have a datapath in between
- Useful
 - Can be used to fix timing violations

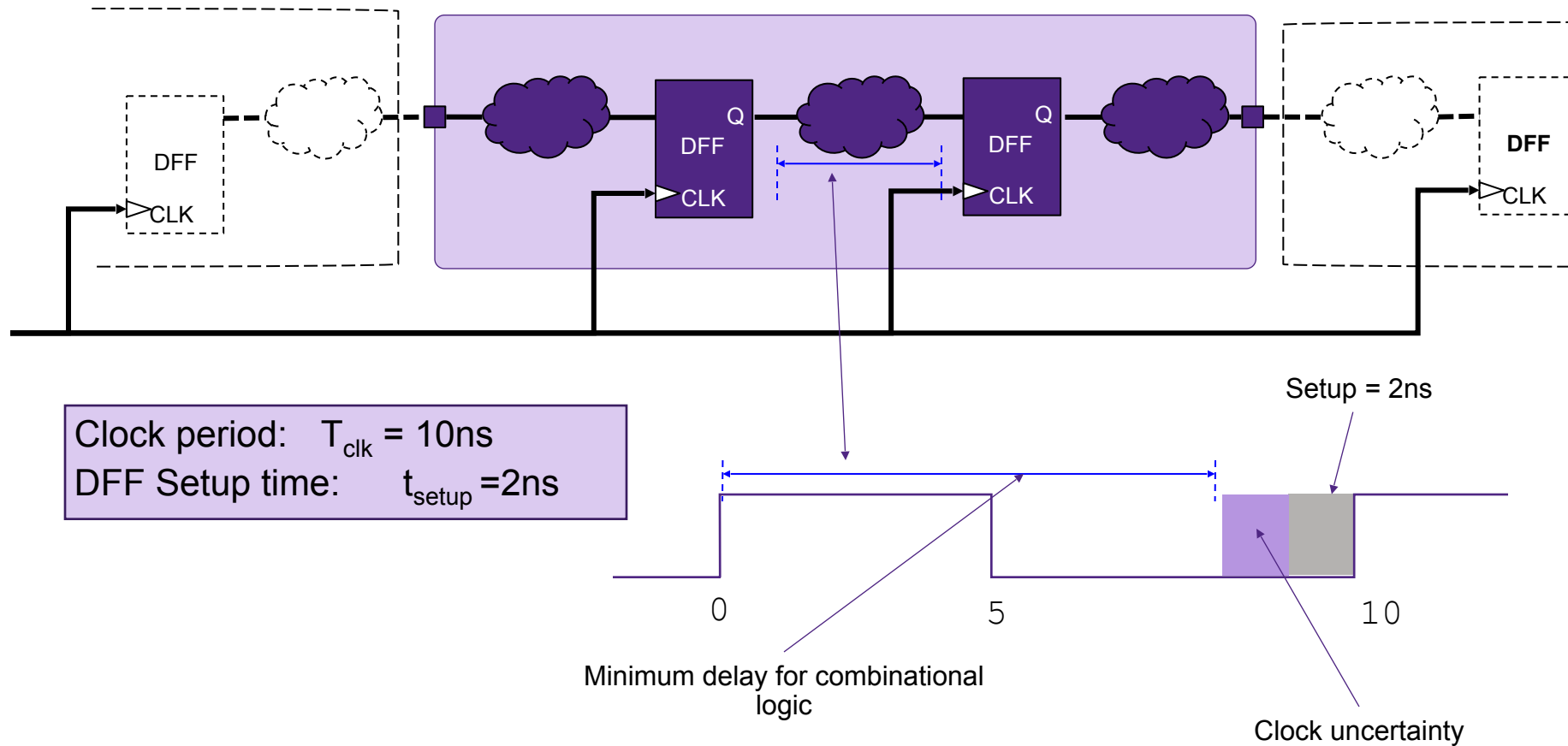
Clock Jitter

Ideal
Reference

Actual
Waveform

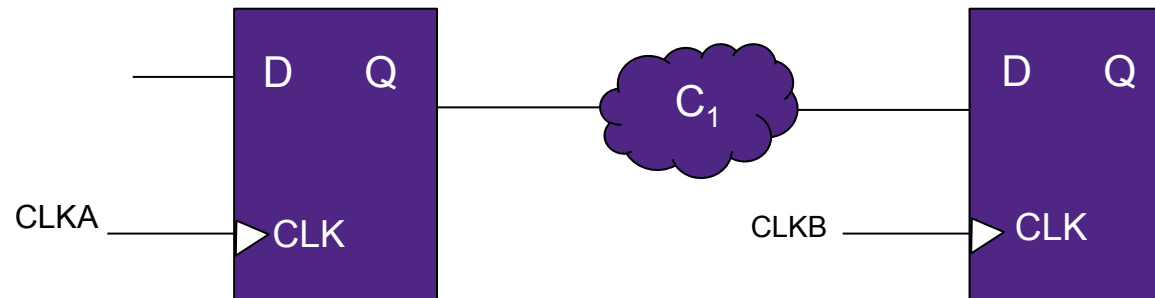


Clock Uncertainty Modeling Example



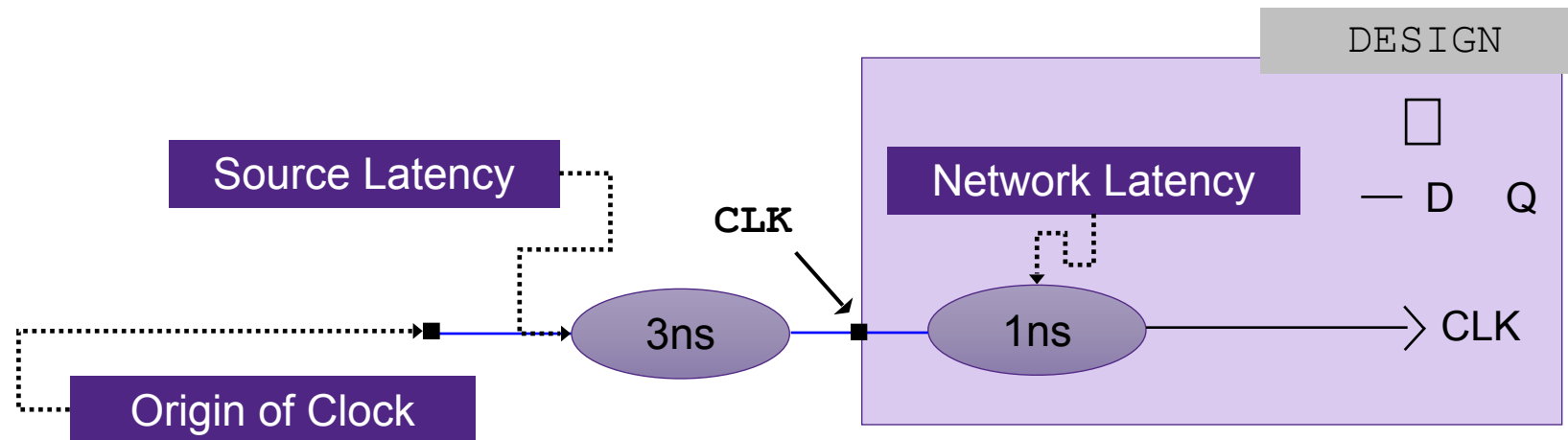
Inter-Clock Uncertainty

```
set_clock_uncertainty -from CLKA -to CLKB -hold 0.3  
set_clock_uncertainty -from CLKA -to CLKB -setup 0.6
```



Modeling Clock Latency

- Clock latency can be of two types:
 - **Source latency:** the delay of clock signal from its source to design's input port (useful when clock generation circuitry is not part of the design)
 - **Network latency:** average propagation delay of clock signal inside the design itself



Clock Latency

```
set_clock_latency 0.9 [get_clocks CLK]
```

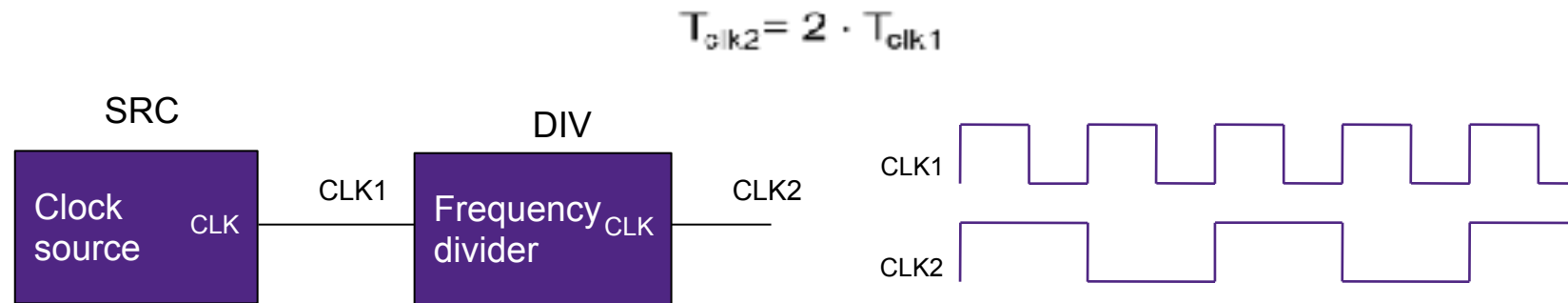
```
set_clock_latency 0.8 -source [get_clocks CLK]
```

```
set_clock_latency 0.7 -source -min /  
[get_clocks CLK]
```

```
set_clock_latency 0.3 -source -max /  
[get_clocks CLK]
```

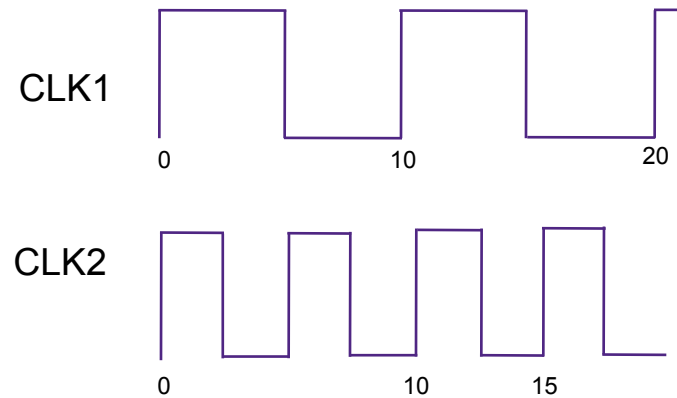

Divided Clock (Generated)

```
create_clock -name CLK1 10 [get_pins SRC/CLK]
create_generated_clock -name CLK2 -source SRC/CLK -divide_by 2 \
    [get_pins DIV/CLK]
```



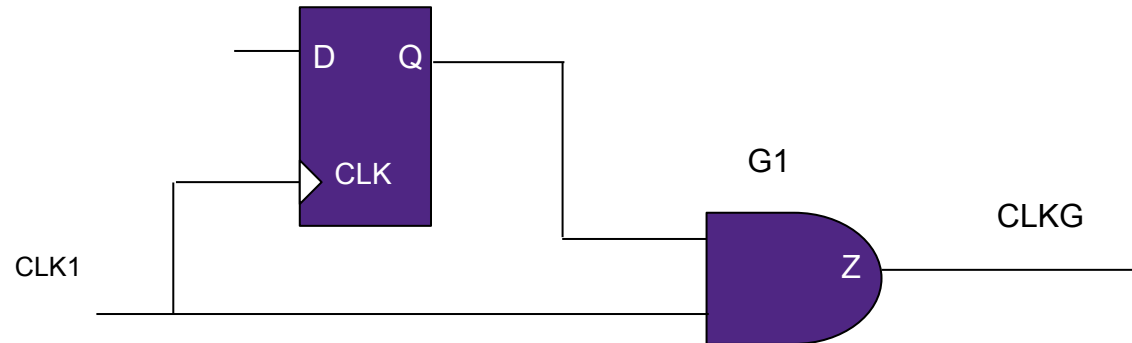
Multiplied Clock (Generated)

```
create_clock -period 10 -waveform {0 5} [get_ports CLK1]  
create_generated_clock -name CLK2 -source [get_ports CLK1] -multiply_by 2 [get_pins MULT/CLK]
```



Gated Clock (Generated)

```
create_clock 10 [get_ports CLK1]  
create_generated_clock -name CLKG -divide_by 1 -source CLK1 [get_pins G1/Z]
```



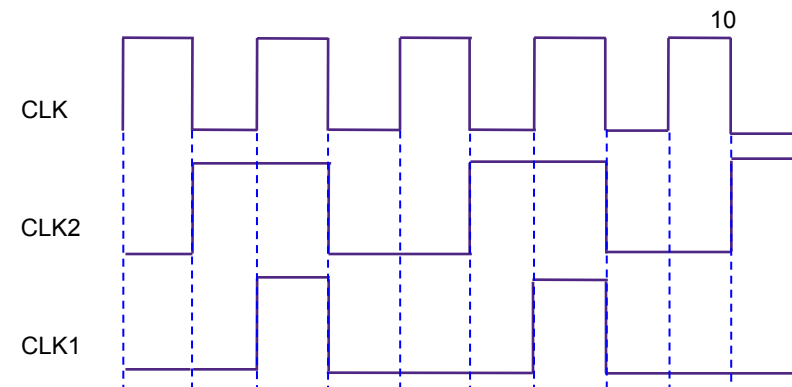
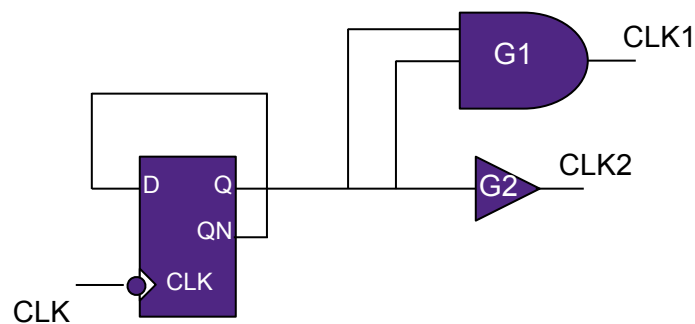
Inverted Clock (Generated)

- Create an inverted generated clock:
 - With respect to the source clock
 - With the same period.

```
create_clock 10 [get_ports CLK1]
create_generated_clock -divide_by 1 -combinational -invert -source CLK1 \
    [get_pins G1/Z]
```

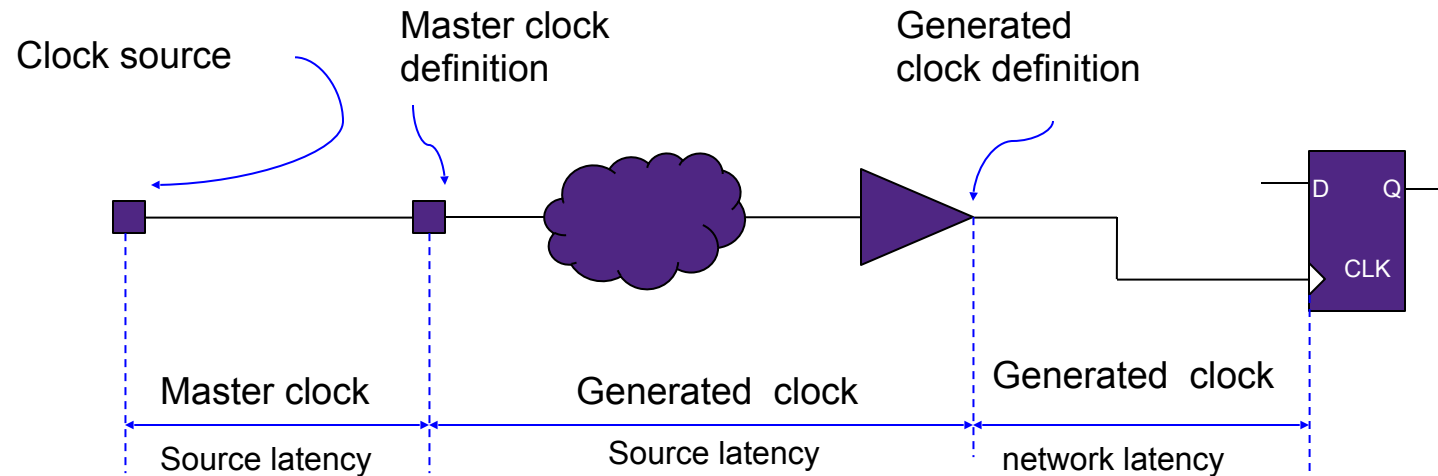
Generated by Edges

```
create_clock 2 [get_ports CLK]
create_generated_clock -name CLK1 -edges {3 4 7} -source DCLK [get_pins G1/Z]
create_generated_clock -name CLK2 -edges {2 4 6} -source DCLK [get_pins G2/Z]
```



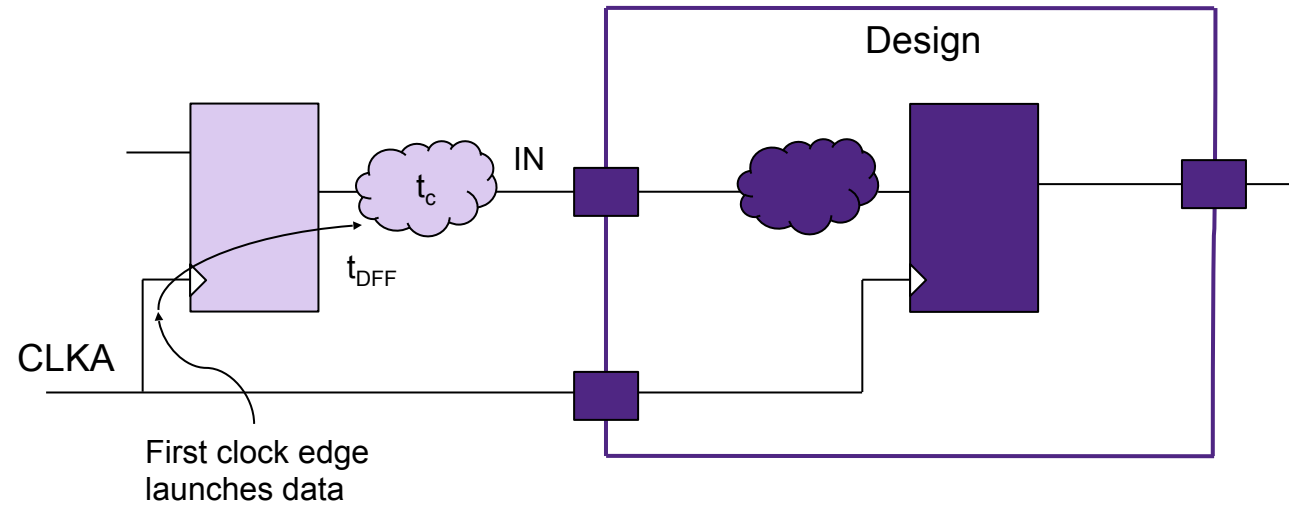
Latency of Generated Clock

- Generated clock total latency is the sum of:
 - Master clock source latency
 - Generated clock source latency
 - Generated clock network latency



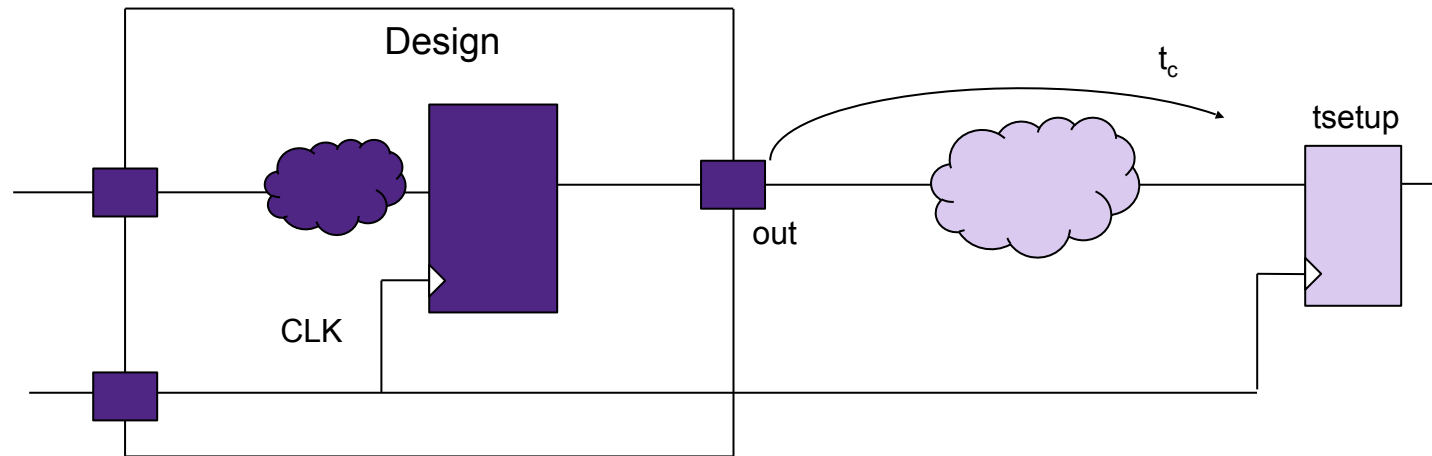
Input Delay

```
set_input_delay -clock CLKA -max [expr TDFE + Tc] [get_ports IN]
```



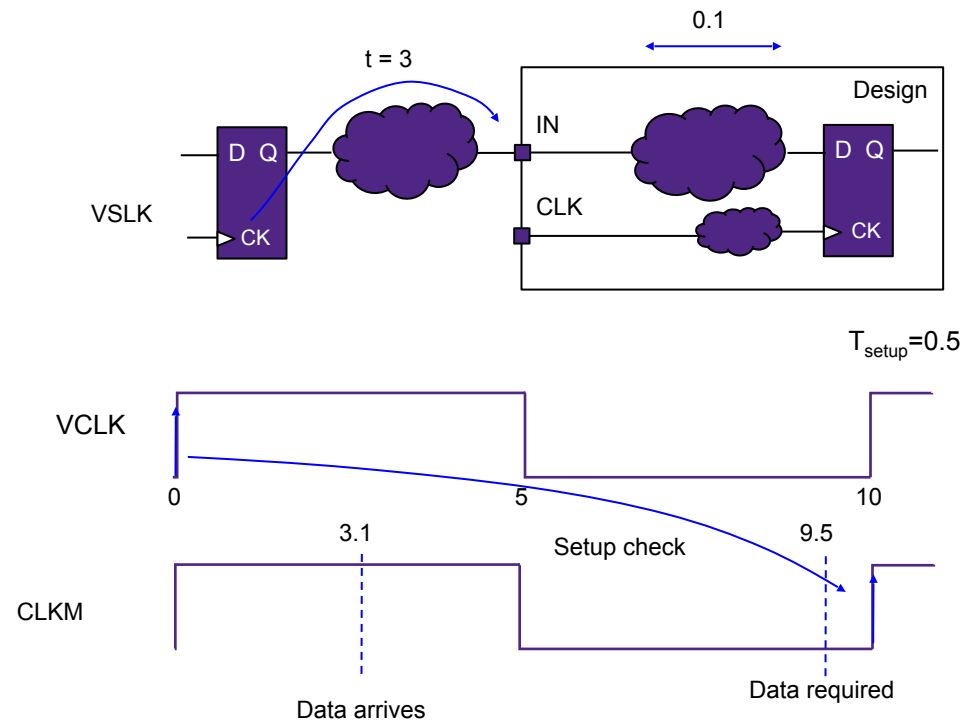
Output Delay

```
set_output_delay -clock CLK -max [exp tc + tsetup] [get_ports OUT]
```

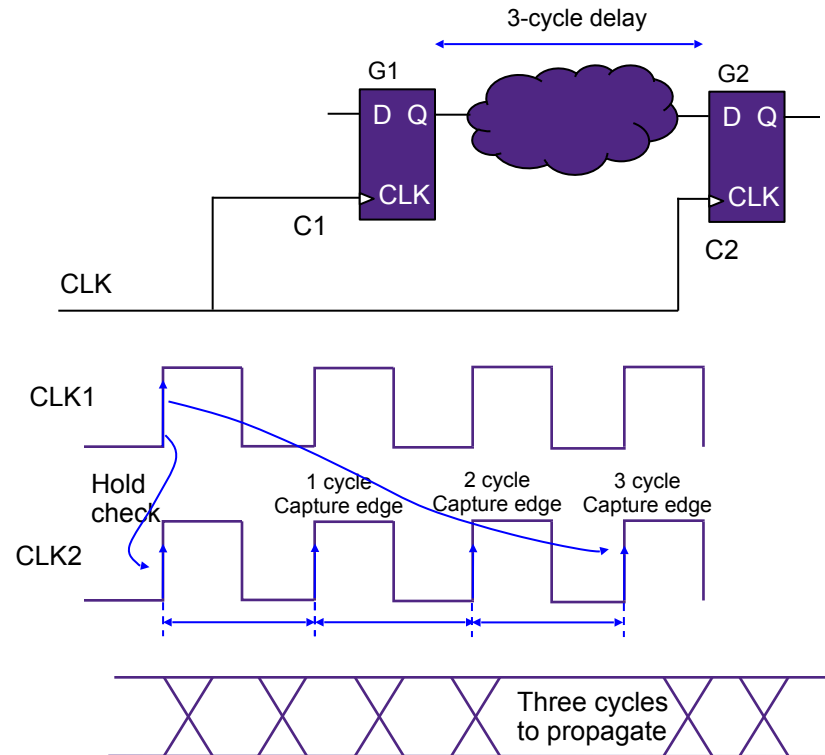


Input Path with Virtual Clock

```
create_clock -name VCLK -period 10 -waveform {0 5}  
set_input_delay -clock VCLK -max 3 [get_ports IN]
```



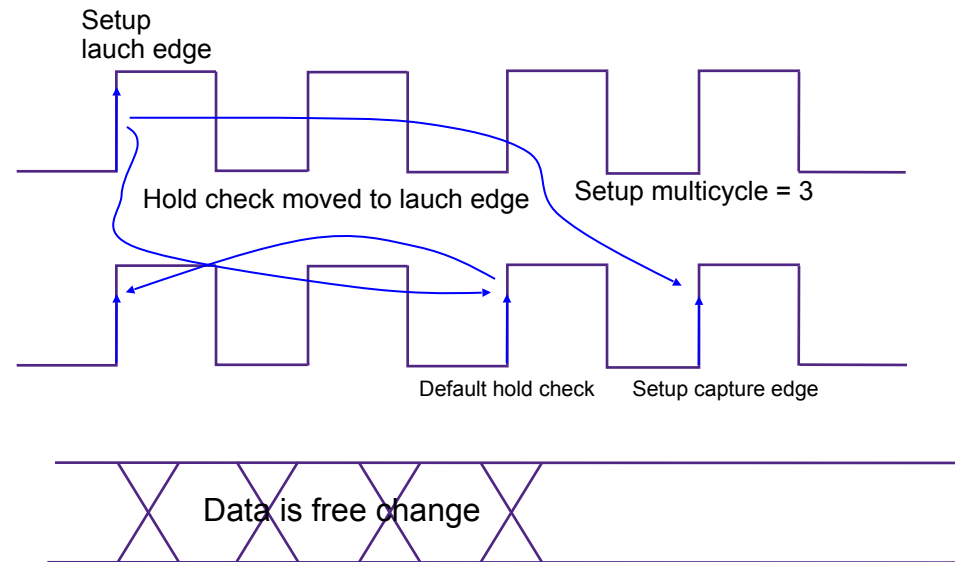
3-cycle Setup Clock



```
create_clock -name CLK -period 10 [get_ports CLK]
set_multicycle_path 3 -setup -from [get_pins G1/Q] -to [get_pins G2/D]
```

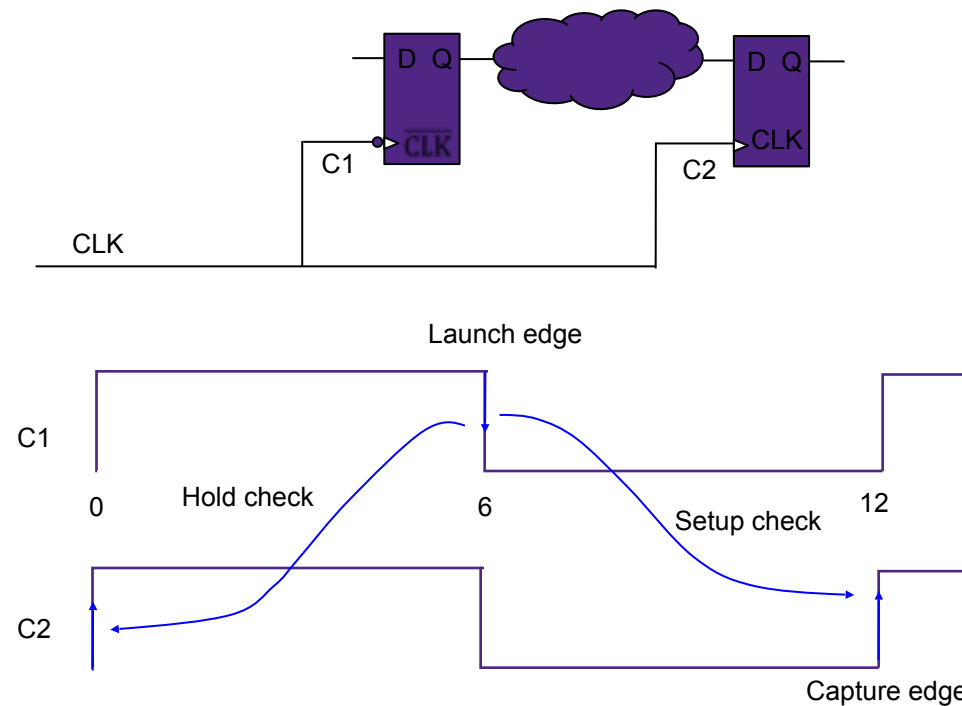
2-cycle Hold Check

```
set_multicycle_path 2 -hold -from [get_pins G1/Q] -to [get_pins G2/D]
```



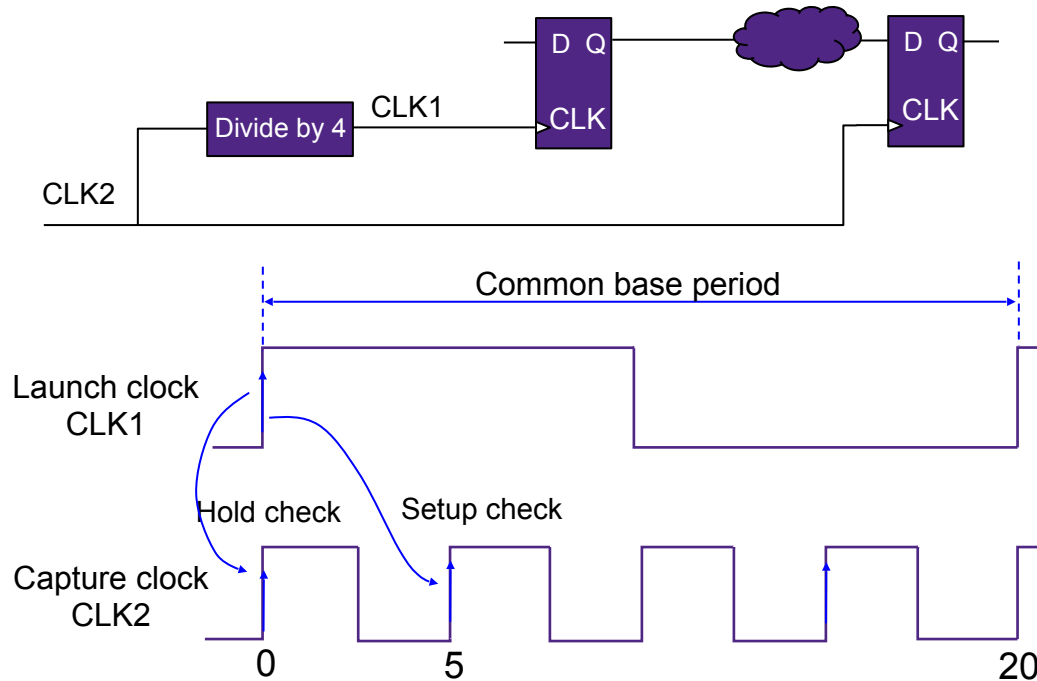
Half-Cycle Path

- The design that has both negative and positive edge triggered flip-flops, can have half-cycle paths



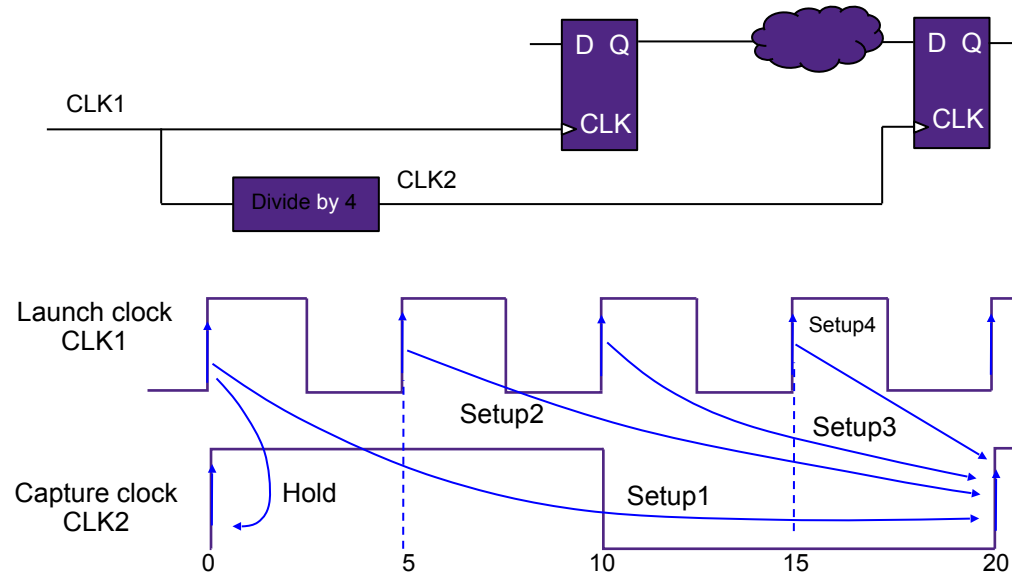
Slow to Fast Clock Domain

```
create_clock -name CLK1 -period 20 -waveform {0 10} [get_ports CLK1]
create_clock -name CLK2 -period 5 -waveform {0 2.5} [get_ports CLK2]
```



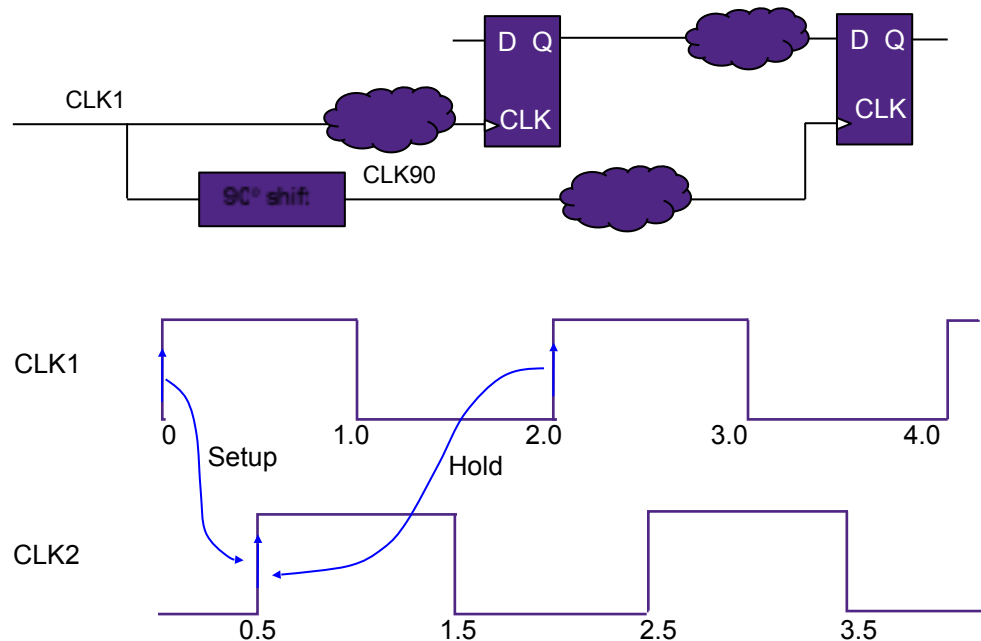
Fast to Slow Clock Domain

```
create_clock -name CLK1 -period 5 -waveform {0 2.5} [get_ports CLK1]  
create_clock -name CLK2 -period 20 -waveform {0 10} [get_ports CLK2]
```



Phase Shifted Multiple Clocks

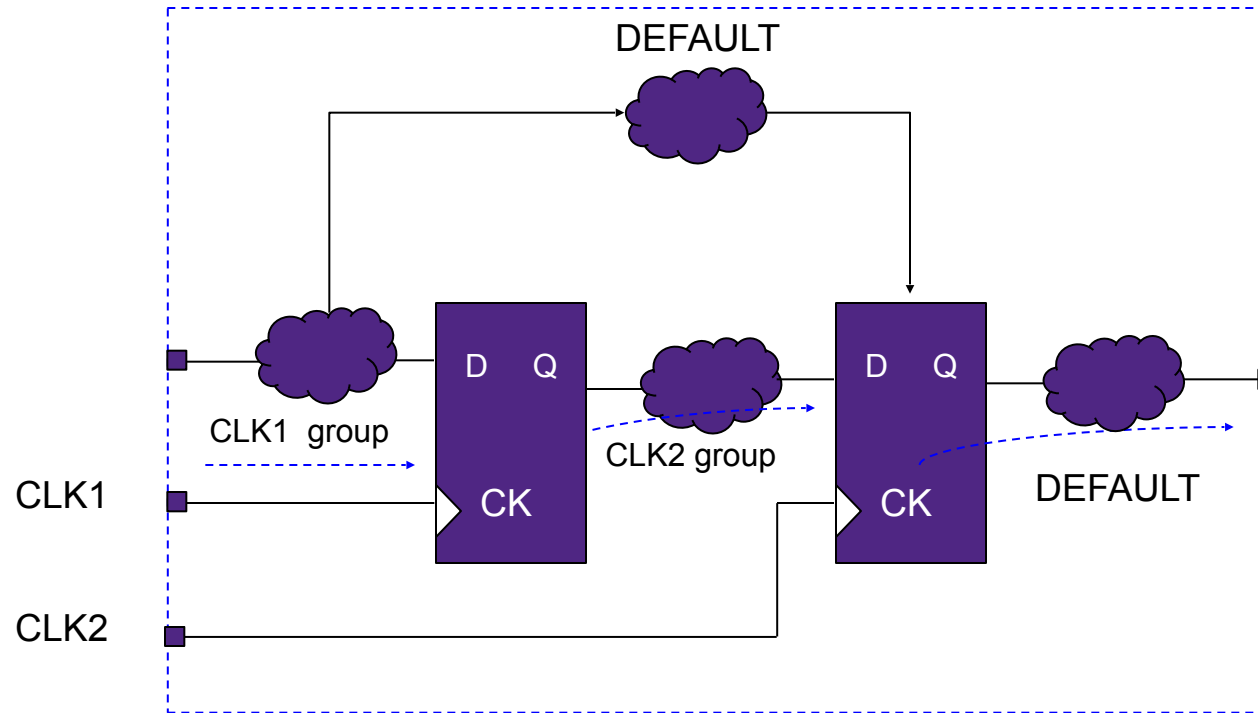
```
create_clock -period 2.0 -waveform {0 1.0} [get_ports CLK1]  
create_clock -period 2.0 -waveform {0.5 1.5} [get_ports CLK90]
```



Path Groups

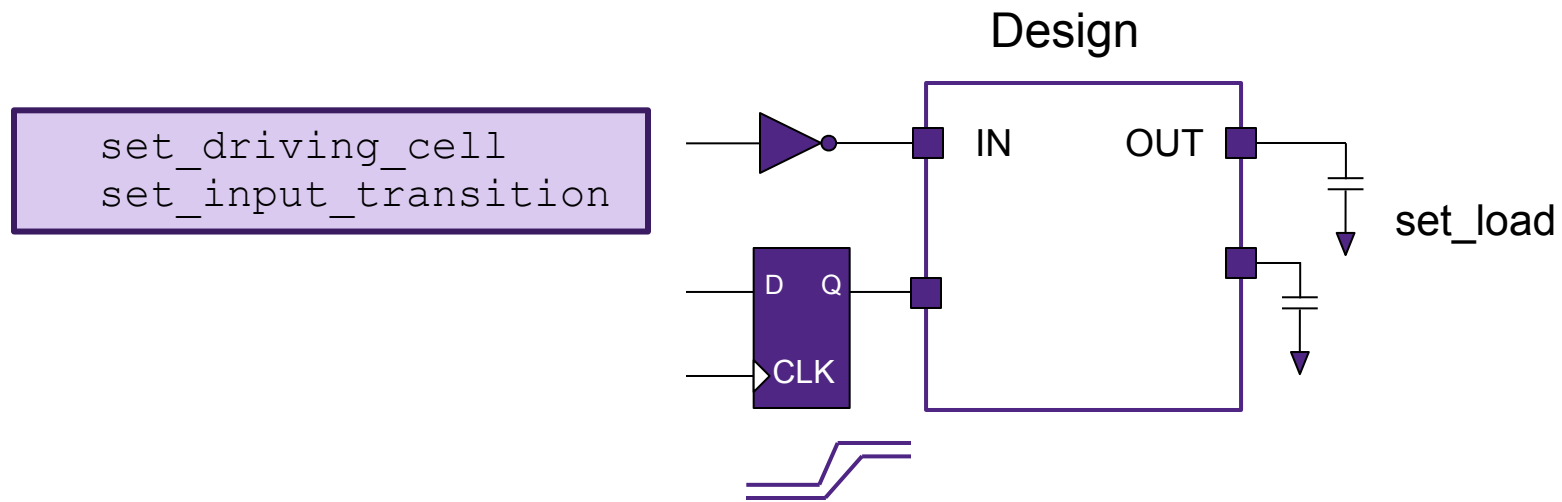
- Path group is created of paths working with the same clock
- Non-clocked paths are in default path group
- The STA is performed on each path group separately

Path Groups (2)



Design Externals

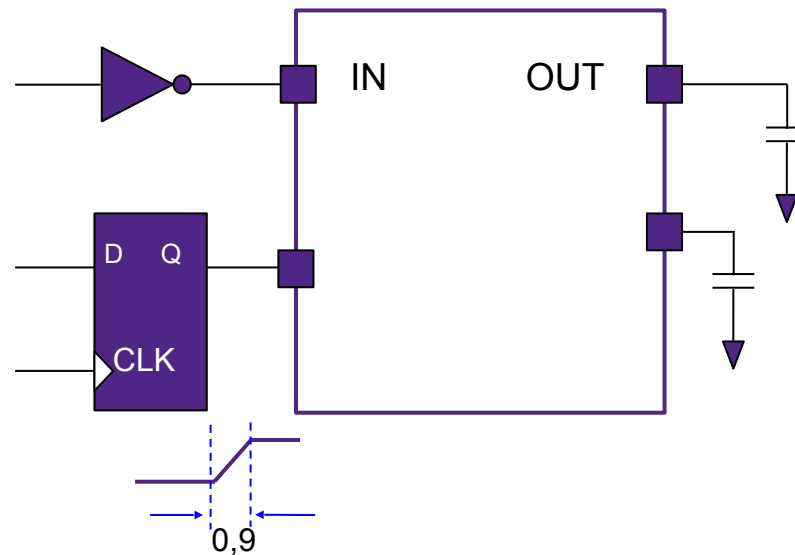
- The following attributes are required to be set:



Driving Cell

```
set_driving_cell -lib_cell INV -library test [get_ports IN]  
set_input_transition 0.9 [get_ports IN]
```

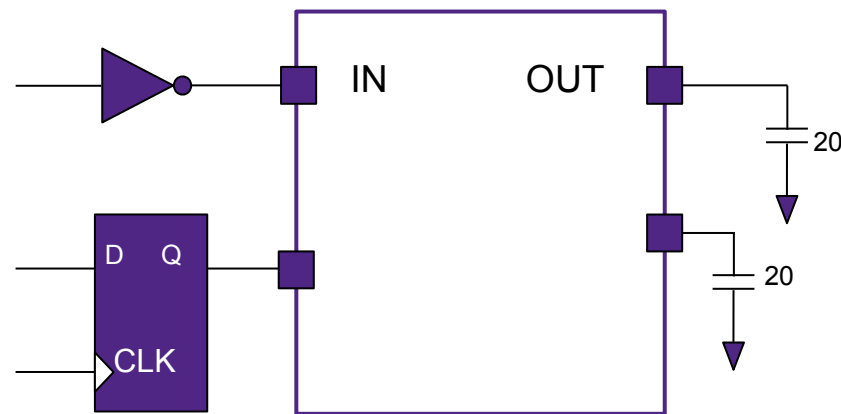
Design



Capacitive Load

```
set_load 20 [all_outputs]
```

Design



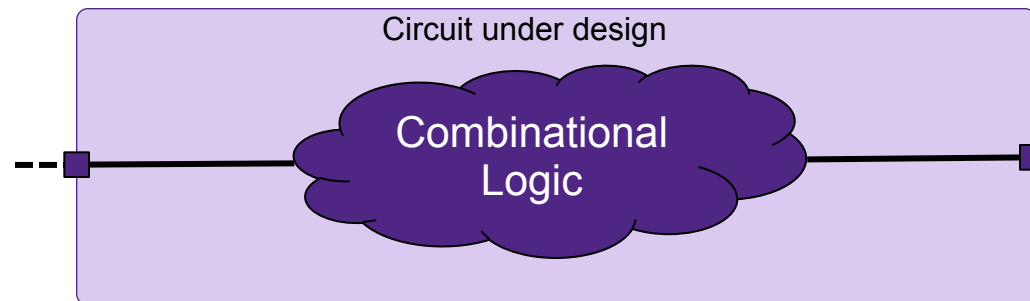
Design Rule Checks

- Rules for all ports and pins in the design to meet transition time and capacitance limits:
 - Maximum transition
 - Maximum capacitance

```
set_max_transition 0.3 [current_design]  
set_max_capacitance 0.7 [current_design]
```

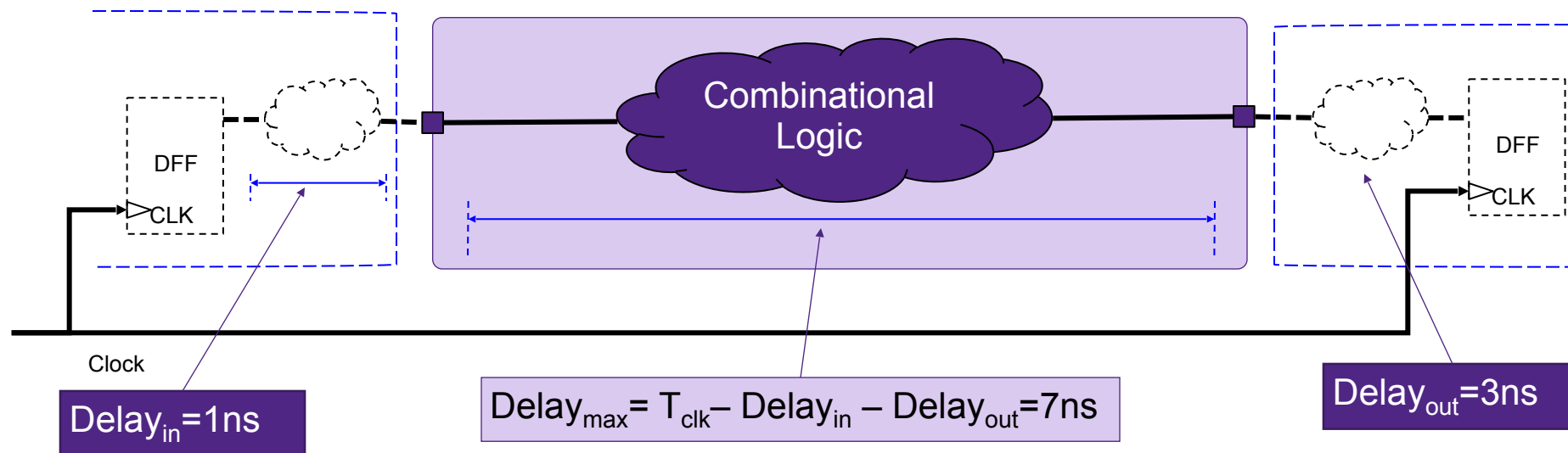
Combinational Designs

- Sometimes the design is a combinational circuit
- No clock to constraint timing



Combinational Designs

- Combinational circuit is put in the same clocked environment as the clocked one
- An abstract clock called “Virtual clock” is defined for this environment
- By setting correct clock period, input/output delays of the combinational logic can be controlled



Refining

- `set_case_analysis`
 - Constant value on a pin of a cell or input port
- `set_disable_timing`
 - Breaks a timing arc
- `set_false_path`
 - Specifies false path not real which implies that these paths are not checked in STA
- `set_multicycle_path`
 - Specifies multi-cycle paths

Point-to-point Specification

```
set_max_delay 4 -to G1
set_max_delay 1 -from G1 -to G2
set_max_delay 3 -from G1 -through G2 -to G3

set_min_delay 3 -from [get_clocks CLK1] -to [get_clocks CLK2]
```

Thank You